9. What is lanthanide contraction and what are the effects of lanthanide contraction?
10. complete the following
    a. $MnO_4^{2-} + H^+ \longrightarrow ?$
    b. $C_6H_5CH_3 \xrightarrow[KMnO_4]{acidified} ?$
    c. $MnO_4^- + Fe^{2+} \longrightarrow ?$
    d. $KMnO_4 \xrightarrow[\text{Red hot}]{\Delta} ?$
    e. $Cr_2O_7^{2-} + I^- + H^+ \longrightarrow ?$
    f. $Na_2Cr_2O_7 + KCl \longrightarrow ?$
11. What are interstitial compounds?
12. Calculate the number of unpaired electrons in Ti^{3+} , Mn^{2+} and calculate the spin only magnetic moment.
13. Write the electronic configuration of Ce^{4+} and Co^{2+}.
14. Explain briefly how +2 states becomes more and more stable in the first half of the first row transition elements with increasing atomic number.
15. Which is more stable? Fe^{3+} or Fe^{2+} - explain.
16. Explain the variation in $E^0_{M^{3+}/M^{2+}}$ 3d series.
17. Compare lanthanides and actinides.
18. Explain why Cr^{2+} is strongly reducing while Mn^{3+} is strongly oxidizing.
19. Compare the ionization enthalpies of first series of the transition elements.
20. Actinoid contraction is greater from element to element than the lanthanoid contraction, why?
21. Out of $Lu(OH)_3$ and $La(OH)_3$ which is more basic and why?
22. Why europium (II) is more stable than Cerium (II)?
23. Why do zirconium and Hafnium exhibit similar properties?
24. Which is stronger reducing agent Cr^{2+} or Fe^{2+}?
25. The $E^0_{M^{2+}/M}$ value for copper is positive. Suggest a possible reason for this.
26. predict which of the following will be coloured in aqueous solution Ti^{2+} , V^{3+} Sc^{4+}, Cu^+ ,Sc^{3+}, Fe^{3+}, Ni^{2+} and Co^{3+}
27. Describe the variable oxidation state of 3d series elements.
28. Which metal in the 3d series exhibits +1 oxidation state most frequently and why?
29. Why first ionization enthalpy of chromium is lower than that of zinc?
30. Transition metals show high melting points why?

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